

TrackAlert Vision

An Autonomous Indoor Patrol Robot with Obstacle Avoidance, Environmental Monitoring & Wireless Safety Alerts

CSE461: Introduction to Robotics | Group 1 | BRAC University | Section 09 | Instructors: RFF, KHR

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MOVE → MONITOR → DETECT → RESPOND → REPORT

Abstract

TrackAlert Vision is a low-cost autonomous patrol robot for controlled indoor spaces such as warehouses, labs, and storerooms. It follows a line to patrol, avoids obstacles with a servo-mounted ultrasonic sensor, and monitors for intruders, fire, gas/smoke, and tip-over. An Arduino Mega 2560 handles real-time sensing and motor control; an ESP32-S3-CAM provides camera and Wi-Fi. On any hazard it stops, sounds a buzzer, shows the event on an LCD, and sends a captured image with an alert to the operator's Telegram.

Introduction

Motivation & Problem

Human patrolling is tiring, costly, and error-prone; fixed cameras only cover fixed spots. An affordable mobile system is needed that patrols on its own and reports only when something is wrong.

Objectives

- Autonomous line-guided patrol of a fixed route
- Obstacle detection, avoidance & route recovery
- Detect intruders (PIR), fire (flame), gas/smoke (MQ-2)
- Detect tip-over (MPU6050) and stop safely
- Send image alerts to a remote operator (Telegram)

Real-World Relevance

Security patrols, warehouse safety, and lab monitoring — less manpower, earlier hazard detection.

System Overview — Concept

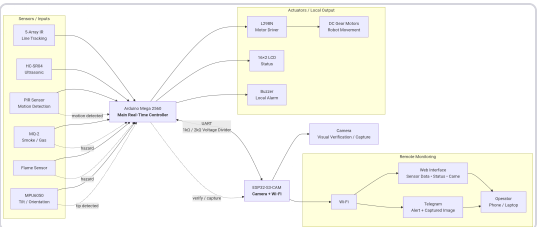


Fig 1. System concept & signal flow

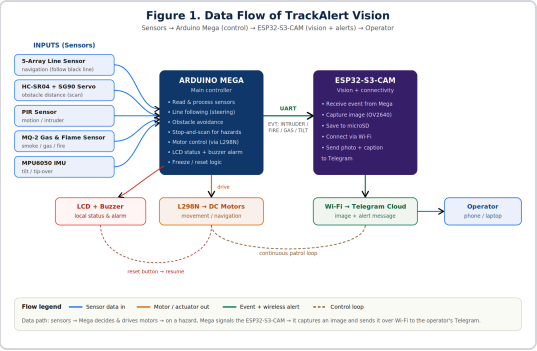


Fig 2. Data flow across the two-controller system

Structural Design (CAD)

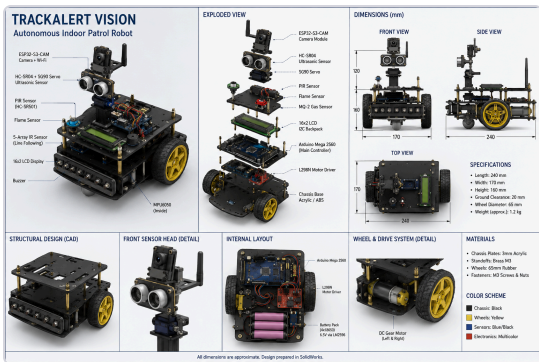
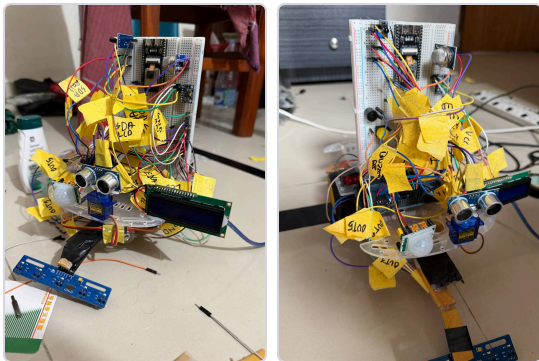


Fig 3. CAD — exploded view, dimensions & detail views (SolidWorks)

Prototype



Assembled robot

Sensor head & wiring

Hardware Components

Arduino Mega 2560	Main real-time controller
ESP32-S3-CAM	Camera, Wi-Fi, Telegram & /capture
5-Array IR Sensor	Line following / navigation
HC-SR04 + SG90	Servo-scanned obstacle detection
PIR (HC-SR501)	Intruder / motion detection
Flame + MQ-2	Fire and gas / smoke sensing
MPU6050	Tilt / tip-over detection (I2C)
L298N + 2 motors	Motor driver & movement
16x2 LCD + Buzzer	Local status & alarm
Power	5V bank + 4x18650 → LM2596 6.5V

Key Specifications

Dimensions	240 × 170 × 160 mm
Weight	~1.2 kg
Wheel dia.	65 mm
Motor supply	6.5 V (LM2596)
Detection	Obstacle 15 cm · PIR ~1 m

Control Circuit Diagram

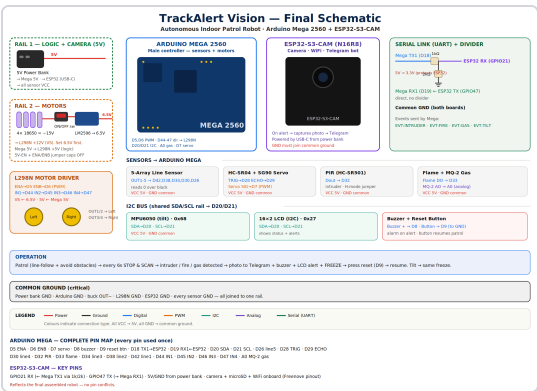


Fig 4. Full schematic — two power rails, common ground, sensors & UART link

Functionalities

Mechanical

Two DC gear motors drive a 2-wheel chassis; a servo sweeps the ultrasonic sensor to find a clear path.

Software / Decision Logic

- Proportional line-following (5-sensor array)
- Obstacle → turn, pass, drive until line re-found
- Periodic 5s stop-and-scan (servo sweep)
- Hazard → alarm + LCD + Telegram photo + freeze/reset

Navigation

Line-guided path with reactive obstacle avoidance — no costly LiDAR/SLAM.

HW–SW Communication

- UART:** Mega ↔ ESP32 (1k/2k divider) — EVT:INTRUDER/FIRE/GAS/TILT
- I2C:** MPU6050 (0x68) + LCD (0x27) on one bus
- Wi-Fi/HTTPS:** ESP32 → Telegram → operator

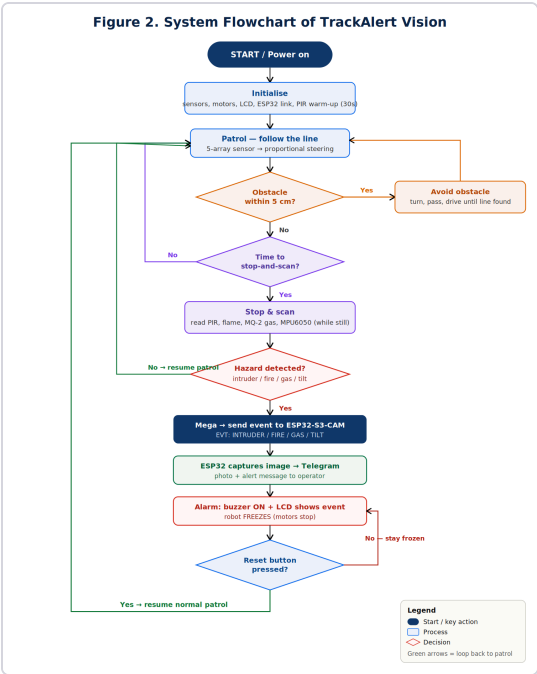


Fig 5. Decision flowchart

Applications & Competitors

Applications

- Warehouse & storeroom security patrols
- Laboratory fire / gas safety monitoring
- Indoor facility surveillance

Vs. Existing Systems

Commercial robots (Knightscope) and research platforms (Microsoft indoor patrol) use LiDAR, thermal imaging, and onboard AI at high cost. TrackAlert Vision is a simpler, educational, low-cost alternative for controlled indoor use.

Advantages

- Very low cost (~BDT 4,486), modular & reusable
- Multi-hazard detection + instant image alerts
- Energy-efficient (rechargeable cells, light compute)

Limitations

- Needs a physical guide line; weak on sharp corners
- Short-range flame sensing; prototype-grade chassis
- PIR read only when stationary

Conclusion

The prototype patrols autonomously, avoids obstacles and recovers its route, detects intruders and hazards, and reports them with a live image — a full move-monitor-detect-respond-report cycle on a low-cost platform.

Future Work

- SLAM/LiDAR navigation (no physical line)
- On-device AI person-detection on the ESP32-S3
- Cloud dashboard & multi-robot coordination
- Industrial-grade sensors & sturdier chassis

References (key)

Santos & Santos (2024). *Telegram: ESP32-CAM take & send photo*. Random Nerd Tutorials. Arduino (n.d.). *Obstacle avoiding robot*. Arduino Project Hub. Lough, B. (n.d.). *Universal Arduino Telegram Bot*. GitHub.